

Course: Digital Image Processing

Subtopic: Image Acquisition

1. Introduction

Definition

Image acquisition is the **first and most fundamental step** in the process of digital image processing.

It refers to the **capturing of an image** from the real-world scene using an image sensor (such as a camera, scanner, or satellite sensor) and converting it into a **digital format** that can be processed by a computer.

In simple terms, image acquisition is the **process of obtaining a digital image for analysis**.

2. Purpose of Image Acquisition

The purpose of image acquisition is to obtain an accurate, high-quality image that represents the physical scene as closely as possible so that further processing (like enhancement, segmentation, or recognition) can be performed effectively.

3. Image Acquisition System Components

An image acquisition system typically consists of the following components:

1. Energy Source

- Provides illumination or radiation that interacts with the object.
- Examples: visible light, X-rays, infrared rays, or microwaves.

2. Object or Scene

- The physical item or environment whose image is to be captured.

3. Imaging Sensor

- Converts reflected or transmitted energy from the object into electrical signals.
- Common types of sensors include:
 - **CCD (Charge-Coupled Device)** cameras
 - **CMOS (Complementary Metal-Oxide Semiconductor)** sensors
 - **Infrared sensors** (for thermal imaging)
 - **Satellite sensors** (for remote sensing)

4. Signal Conversion Hardware

- Converts the **analog signal** from the sensor into a **digital form** using an **ADC (Analog-to-Digital Converter)**.

5. Computer or Processor

- Stores the digital image and prepares it for processing or analysis.

6. Display or Storage Device

- Used for visualizing or saving the acquired image.
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4. Image Acquisition Methods

There are several methods of image acquisition depending on the application and type of image required.

A. Single Sensor System

- Uses one sensor (camera or photodiode array).
- Common in webcams, scanners, and basic cameras.
- Produces a 2D image directly.

B. Line-Scan (Pushbroom) System

- Captures one line (row of pixels) at a time as the object or sensor moves.
- Used in industrial inspection and satellite imaging.
- Suitable for continuous motion environments (e.g., conveyor belts).

C. Array Sensor System

- Uses a matrix of sensors arranged in rows and columns.
- Captures full 2D images in a single exposure.
- Common in digital cameras and smartphones.

D. Multispectral and Hyperspectral Imaging

- Captures images in multiple spectral bands beyond the visible range (infrared, ultraviolet).
- Used in remote sensing, agriculture, and medical applications.

E. 3D Image Acquisition

- Uses **depth sensors**, **stereo cameras**, or **LiDAR** to capture 3D information about a scene.
 - Produces both intensity and depth data.
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5. Image Acquisition Geometry

The geometry of image acquisition describes the **spatial relationship** between the imaging device, the object, and the illumination source.

Types of Geometries

1. Reflection Geometry

- Light is reflected from the object into the sensor.
- Common in visible light photography.

2. Transmission Geometry

- Light passes through the object before reaching the sensor.
- Used in X-ray imaging and microscopy.

3. Scattering Geometry

- Used when light scattering properties are analyzed, such as in medical and biological imaging.
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6. Digital Image Formation

Once light energy reaches the sensor, it undergoes several steps before forming a digital image:

1. **Sensing:** Sensor detects energy reflected or transmitted from the scene.
2. **Analog Signal Generation:** Sensor outputs a continuous electrical signal.
3. **Sampling:** The continuous signal is sampled into discrete values (pixels).
4. **Quantization:** Each pixel's brightness (intensity) is assigned a numeric value.
5. **Digitization:** The sampled and quantized data are converted into a digital image matrix that can be stored and processed.

Mathematically: A digital image can be represented as

$f(x, y)$

- → spatial coordinates of pixels
 - → intensity (brightness) at point (x, y)
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7. Factors Affecting Image Acquisition Quality

1. Lighting Conditions

- Poor lighting can cause shadows, glare, or noise.

2. Sensor Resolution

- Determines the detail captured in an image.

3. Lens Quality

- Impacts focus, distortion, and color accuracy.

4. Sampling Rate

- Higher sampling provides better spatial detail.

5. Environmental Factors

- Temperature, vibration, and humidity may affect image clarity.

6. Noise

- Random variations in brightness or color introduced during acquisition.
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8. Pre-Acquisition and Post-Acquisition Steps

Pre-Acquisition

- Adjusting illumination, camera angle, and focus.
- Calibrating sensors for color and exposure.

Post-Acquisition

- Converting raw image data to standard formats (e.g., JPEG, TIFF, BMP).
 - Performing **image preprocessing** (denoising, normalization).
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9. Importance of Image Acquisition in Digital Image Processing

- It defines the **starting point** for all subsequent processing operations.
 - Poor image acquisition leads to degraded performance in later stages such as enhancement, segmentation, or recognition.
 - Ensures the **accuracy and reliability** of image analysis applications.
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10. Summary

- **Image acquisition** is the process of **capturing and digitizing images** for processing.
 - It involves **sensors, converters, and storage systems** working together to create a digital representation of the real world.
 - The **quality of image acquisition** directly influences the success of later image processing stages.
 - Proper selection of **sensors, geometry, and lighting** ensures high-quality data for accurate analysis and recognition.
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